

Research Idea: J0604

ALMA Thesis Planner (automated pipeline)

Title: Do the ground-state oxygen-hydride absorption lines trace one absorbing layer or two? A redshift-free differential study of H₂O+ (and OH+ where covered) in the 40-source z=2-5 starburst sample, ALMA 2023.1.00804.S

Core question. The PI's science case turns on whether the molecular-ion absorption in these starbursts arises in outflowing/infalling gas or in the layers immediately around the active star-forming region. This thesis attacks that question with a quantity that requires no systemic redshift at all: within a single galaxy, do two different ground-state hydride absorption features sit at the *same* velocity, with the same width and the same relative strength? A systematic, nonzero offset between them means the two ions occupy kinematically distinct layers — the ionized, turbulent, likely moving component is separable from the denser one — and a null result, quoted at a stated precision, is an equally publishable constraint. Because both features are measured from observed sky frequencies in the same headers of the same delivery, any error in an adopted redshift shifts both by the same velocity and cancels exactly in the difference.

Data. Strictly and only the delivered products of 2023.1.00804.S: calibrated measurement sets, pipeline continuum images and cubes, and the frequency axes in those products' own headers. No external survey catalog, no external redshift catalog, no prior OH+ measurements. The only non-project inputs are the laboratory rest frequencies of the relevant ground-state transitions, taken from a standard molecular spectroscopy line list — these are physical constants, not survey data, and the student records the exact tabulated values and their source in the thesis.

Provenance note, stated up front. The data description quotes *no* numeric frequency, wavelength, or redshift for any line or any target, and says nothing about rest- vs. observed-frame for any value. Nothing numeric is therefore imported from it. Which transitions are actually covered — the two fine-structure components of the H₂O+ ground state, and whether any OH+ ground-state transition also falls in a delivered spectral window — is determined empirically in Week 1 by comparing the tabulated rest frequencies, redshifted by each target's adopted tuning redshift, against the delivered spectral-window coverage in the headers. Every adopted redshift is treated as a provisional tuning value with no error bar attached and is used only to *locate* search windows, never to convert an absolute velocity into an astrophysical claim.

Method (one semester to one year).

Week 0 — access check with a real fallback. Confirm retrievability of the calibrated products for all ~40 targets. If any subset is still proprietary, the sample is whatever is genuinely in hand, and the pipeline (steps 2-4) is built and validated end-to-end on synthetic cubes generated from the delivery's own header parameters, then applied when the data release.

1. *Uniform continuum photometry (weeks 1-5).* Fixed aperture prescription (curve-of-growth to a stated enclosed-flux criterion, unresolved-source fallback), bootstrap uncertainties from residual

images, consistent beam treatment. Deliverable: the sample’s continuum brightness distribution and resolved/unresolved classification — the denominator for every optical depth.

2. *Uniform spectral extraction (weeks 5-11)*. Continuum subtraction and continuum-normalized spectra per source over every delivered spectral window, on a common channel width. Image-plane vs. uv-plane continuum subtraction tested on two representative sources, one choice fixed sample-wide, the difference carried as a documented systematic.

3. *Matched-filter feature search (weeks 11-17)*. Filter bank spanning ~ 50 -800 km/s FWHM, searching absorption and emission. False-alarm rate calibrated empirically on line-free regions and off-source apertures in the same cubes, with candidate emission features masked first so that real sideband emission in these bright starbursts does not inflate the threshold and silently destroy absorption sensitivity. Output per source and per transition: peak optical depth, equivalent width, FWHM, and observed-frequency centroid with uncertainties.

4. *Injection-recovery, image plane only (weeks 17-24)*. Synthetic absorption profiles of known peak optical depth, width, and offset injected into the real continuum-normalized spectra and re-searched blind, yielding a completeness surface per source and the τ recoverable at 90% completeness for a fiducial 200 km/s feature. Visibility-plane injection is deliberately cut as an unbounded CASA compute burden; imaging-stage losses are instead bounded analytically and spot-checked on two or three resolved sources.

5. *The differential measurement — the science (weeks 24-33)*. For every source in which two hydride transitions are both covered and both measured, report (a) the velocity difference between their centroids, computed from the two observed sky frequencies and the two tabulated rest frequencies, with the redshift-cancellation stated explicitly; (b) the ratio of equivalent widths; (c) the ratio of fitted widths. Then test the ensemble for a nonzero mean offset against a permutation null. Where individual sources are too faint, stack each transition separately across the sample: since a redshift error displaces both lines by the *same* velocity, the two stacks are smeared by the same kernel and the difference of their centroids survives — this is the low-S/N route to the same result, and its residual bias is quantified by injection.

6. *Safe floor and write-up (weeks 33-38)*. Per-source detection or completeness-calibrated τ upper limit; incidence rate evaluated only within the subsample whose completeness surfaces reach a common fiducial threshold, so shallow and deep pointings are never averaged together; a permutation test for whether incidence still varies with continuum brightness at matched sensitivity. Deliver the feature catalog, completeness surfaces, and scripts.

Scope limits. No absolute outflow/infall/systemic classification — that needs a systemic redshift the delivery does not supply. No abundances, column densities, fractional ionization, or mass/momentum flux rates: these need covering fraction, geometry, and an excitation model. OH+ enters as a stated sample-selection property with no numbers attached unless one of its transitions is independently found in this project’s own spectral windows.

Contingency. If a second transition is covered in too few sources to support step 5, the thesis reports the sensitivity-matched incidence census and completeness-calibrated τ limits alone, which is already a complete and useful thesis for a 40-source absorption survey.